Doctor–Hospital Assignment Project Plan

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1. Assumptions and Mathematical Formulation

We assume there are N doctors and K hospitals. Each doctor ranks all hospitals from most to least preferred, and hospitals do not have preferences over doctors. Each doctor must be assigned to one hospital. Hospital j has capacity C_j, and we assume the total capacity is large enough for all doctors.

We use the doctor's ranking as the assignment cost. Let c_{ij} be the rank of hospital j for doctor i, where 1 is the first choice. Let $x_{ij} = 1$ if doctor i is assigned to hospital j, and 0 otherwise. Our goal is

$$\min \sum_{i=1}^{N} \sum_{j=1}^{K} c_{ij} x_{ij},$$

subject to

$$\sum_{j=1}^{K} x_{ij} = 1 \quad \forall i, \qquad \sum_{i=1}^{N} x_{ij} \leq C_j \quad \forall j, \qquad x_{ij} \in \{0, 1\}.$$

We will mainly evaluate the result using the average rank of the assigned hospitals. A lower average rank means that doctors are generally assigned closer to their preferred hospitals.

2. Algorithm and Baseline

We plan to use the Hungarian algorithm as our main method. Since one hospital can accept multiple doctors, we will split each hospital into a number of identical slots based on its capacity. For example, a hospital with capacity 3 will be represented by three slots. We can then build a cost matrix using the doctors' rankings and apply the Hungarian algorithm to find an assignment with the minimum total rank cost.

For comparison, we will use a simple greedy method as our baseline. We will go through doctors one at a time and assign each doctor to their highest-ranked hospital that still has space. We will compare the two methods using average rank, total rank cost, and the percentage of doctors who receive their first choice.

3. Relevant References

1. Kuhn, H. W. (1955). "The Hungarian Method for the Assignment Problem." *Naval Research Logistics Quarterly*, 2(1–2), 83–97.
2. Wu, Y., et al. (2025). "Improving Residency Matching Through Computational Optimization." *JAMA Network Open*, 8(6), e2517077.
3. Kuhn, H. W. (1956). "Variants of the Hungarian Method for Assignment Problems." *Naval Research Logistics Quarterly*, 3(4), 253–258.
4. Munkres, J. (1957). "Algorithms for the Assignment and Transportation Problems." *Journal of the Society for Industrial and Applied Mathematics*, 5(1), 32–38.

4. Limitations and Possible Extensions

One limitation is that We assume that every doctor ranks every hospital and that hospitals have no preferences over doctors. Other possible limitations may be that doctors should add all hospitals to the rank. Possible extensions include adding hospital preferences, allowing incomplete preference lists, and adding other constraints, such as specialty, location, and eligibility. Other extensions are using nonlinear preference cost and adding fairness objective, to ensure that everyone's choices are as close to the front as possible.